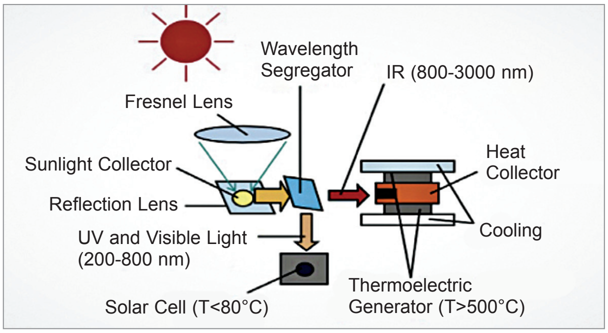

Fig. 14: Arrangement of hybrid PV/TEG system using spectral splitting

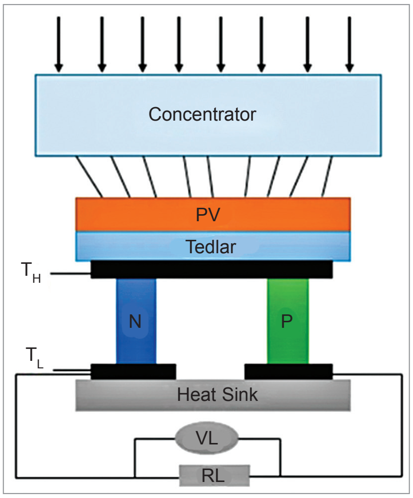

Fig. 15: Hybrid PV/TEG system without reflective component

late the TE legs to suppress lateral heat leaks that degrade thermal efficiency. Encapsulation of thermoelectric legs with aerogels prolongs the life of TE devices.

The primary cause of deterioration of most thermoelectric materials is thermal decomposition or sublimation at high operating temperatures. For example, aerogel present near the surface of skutterudite material, such as $CoSb_3$, prevents transport of Sb vapour by establishing a highly localised equilibrium Sb-vapour atmosphere at the surface of skutterudite.

Some solar absorbers are painted with black paint to increase the heat absorption and are fabricated from metal dielectric multilayer cermet composites that are capable of withstanding more than 950°C.

Photovoltaic/thermoelectric hybrid system

Solar light and its thermal energy can provide sufficient electricity to meet the global energy demand. The range of wavelengths that photovoltaic materials generally use to convert into electricity is between 400nm and 1200nm, the ultra-violet (UV) and visible range. Excess solar radiation is wasted as heat, which decreases the efficiency of PV cells and lowers their life.

TEGs are bidirectional devices that act as heat engines, converting the excess heat into electricity through the thermoelectric effect. Thermoelectric devices utilise the IR region of sunlight to generate electricity and reduce the amount of heat that PV cells dissipate. It is possible to combine PV cells and TEGs to make a hybrid system that can generate more energy. The overall power output of this system would be the sum of the power output from the PV module and the TEG.

The hybrid systems generally follow two configurations—with or without reflective components.

With reflective arrangement the spectral splitting method splits the solar spectrum into two bands. The spectrum below the 800nm cut-off wavelength gets transmitted to the PV module and above 800nm to the TEG. This system has a reflective component called wavelength segregator or prism, where the PV module and the TEG are installed perpendicular to each other. When sunlight passes through the prism, a part of the sunlight is reflected at cut-off wavelength of approximately 800nm and is absorbed by the solar cell. The radiation that is longer than the cut-off wavelength—above 800nm—is reflected to the TEG, as shown in Fig. 14.

The PV module and the TEG convert solar energy into electricity independently. A cooling system is installed on the TEG to maintain the temperature difference. A concentrator increases the light intensity and hence PV modules of reduced surface area can be installed, which results in a reduced installation and maintenance cost. The spectrum splitter allows for low operating temperature and hence maximising the conversion efficiency.

Thermoelectric legs have different values of ZT at different temperatures. Hence the TE material selected depends on the operating temperature. Generally, bismuth telluride is used under 500K, lead telluride between 500-900K, and germanium silicon above 900K operating temperatures.

In configurations without reflective component the PV module is placed as upper component and TEG device as lower component, as shown in Fig. 15. When sunlight falls on it, the PV device absorbs the UV and visible light and rest of the solar spectrum passes through the PV module to the underlying TEG. The IR radiation heats up the TEG top side creating a temperature difference with the cold side. The solar module is coated with tedlar PVF film that offers the best protection against UV, thermal, moisture, chemical and mechanical stress.

Combination of PV and TE generators can make a very efficient device for solar energy utilisation. A PV/TE hybrid system with high concentration ratio and using multijunction PV and Bi2Te3 thermoelectric legs has conversion efficiency of about 32%. The direct electrical contribution of the TEG to the hybrid system's efficiency is enhanced by increasing the Sun's concentration by about 300 times. Even higher efficiency and power